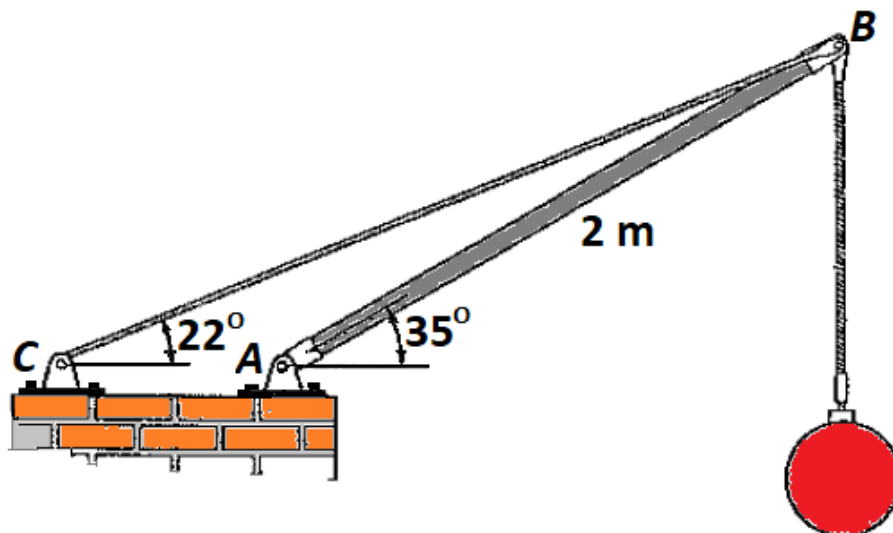
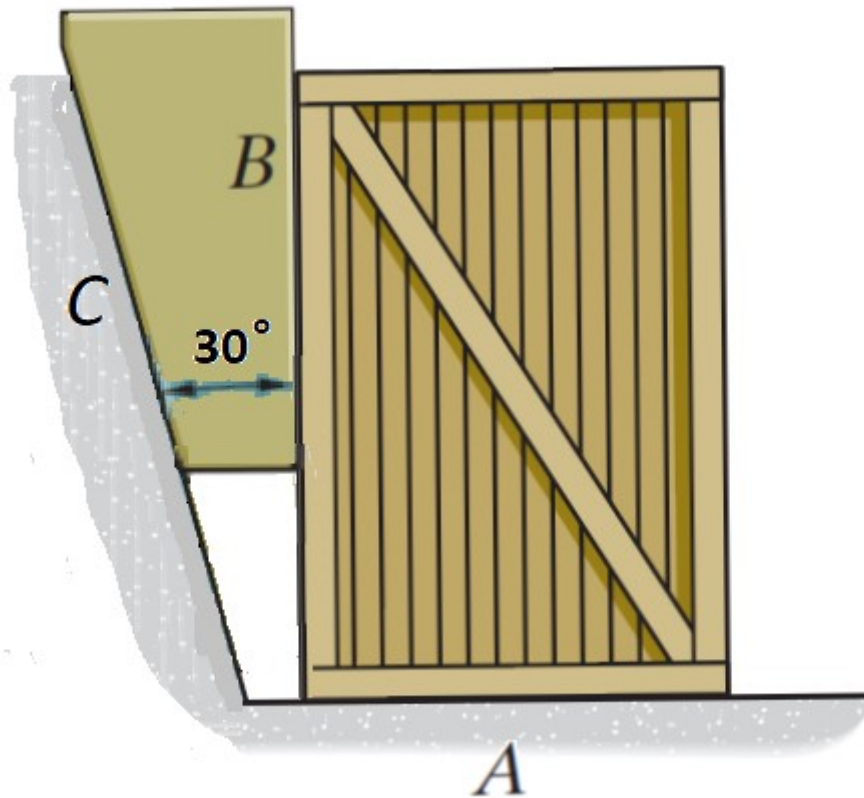


2020-2021 Fall Semester
MTM3701 MECHANICS 1 (Gr.2)
Midterm Exam 1
(23.11.2020)

Q-1. (10 Points) Determine the magnitude of force at the pin A and in the cable BC needed to support the 2000N load. Neglect the weight of the boom AB .



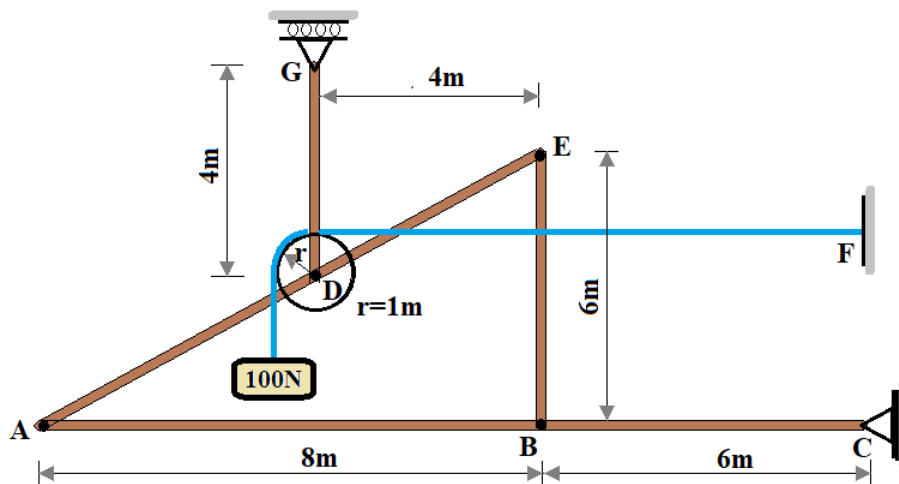
Q-2. (30 Points) Determine the largest weight of the wedge that can be placed between the 8N block and the wall without upsetting equilibrium. The coefficient of static friction at A and C is $\mu_s = 0.5$ and at B $\mu'_s = 0.6$.



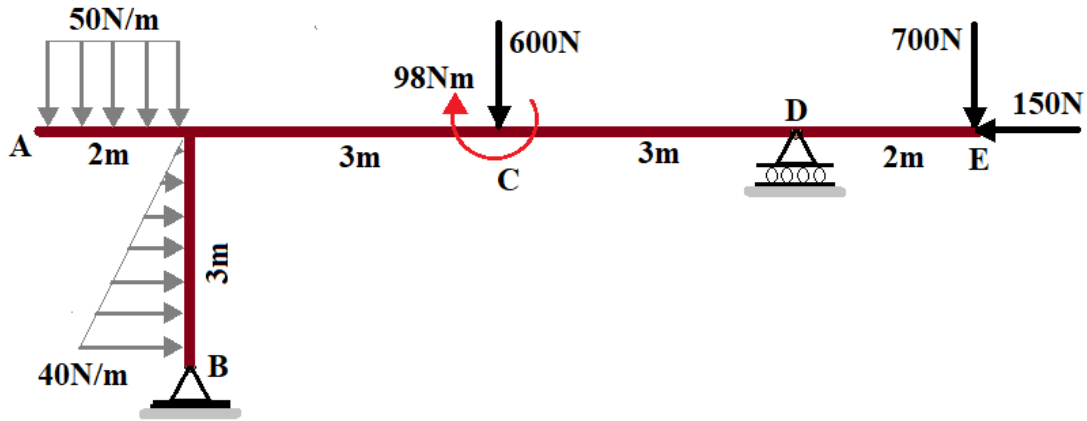
Q-3. (30 Points) The frame given in the figure is attached with a fixed support from the C point and the roller support from the G point. In addition, the F end of the cable passing around the frictionless pulley with radius $r = 1\text{m}$ at point D is horizontally attached to the vertical wall on the right, and a load with a weight of 100N is hung on the other end. Find

- support reactions at support points C and G
- joint reactions at points A, B, D and E, neglecting the friction on the contact surfaces and the weights of all the members in the frame.

Note: DG and EB bars are vertical, ABC bar is in horizontal direction.



Q-4. (30 Points) Find the reaction forces at fixed support and roller support at points B and D, respectively, in the carrier system whose geometry and loading state are given.



Note: Answers will be resolved on A4 paper cleanly and neatly, Name, Surname and number will be written in the upper left corner of the first page.

Duration: 100 min.